

Filter with Numbers and Dates in SQL

Project Description

This lab activity focused on applying numeric and date/time operators in SQL to investigate a simulated security incident. Working within the **organization** database on MariaDB, queries were executed against the **log_in_attempts** table to retrieve login events filtered by date ranges, time windows, and event ID ranges. The operators used include `=`, `>`, `<`, `>=`, `<=`, `<>`, and the **BETWEEN ... AND** clause.

Task 1 — Retrieve Login Attempts After a Certain Date

The investigation required gathering login attempts made after **2022-05-09**. Two queries were run: the first used the strict greater-than operator (`>`) to exclude that date, returning **125 rows**. The second used greater-than-or-equal (`>=`) to include 2022-05-09 itself, expanding the result to **165 rows**.

Logins Strictly After 2022-05-09 (> operator)

```
SELECT *  
FROM log_in_attempts  
WHERE login_date > '2022-05-09';
```

```
clear
MariaDB [organization]> SELECT *
->
-> FROM log_in_attempts
->
-> WHERE login_date > '2022-05-09';
```

event_id	username	login_date	login_time	country	ip_address	success
2	apatel	2022-05-10	20:27:27	CAN	192.168.205.12	0
5	jrafael	2022-05-11	03:05:59	CANADA	192.168.86.232	0
6	arutley	2022-05-12	17:00:59	MEXICO	192.168.3.24	0
7	eraab	2022-05-11	01:45:14	CAN	192.168.170.243	1
9	yappiah	2022-05-11	13:47:29	MEX	192.168.59.136	1
10	jrafael	2022-05-12	09:33:19	CANADA	192.168.228.221	0
11	sgilmore	2022-05-11	10:16:29	CANADA	192.168.140.81	0
13	mrhah	2022-05-11	09:29:34	USA	192.168.246.135	1
14	sbaelish	2022-05-10	10:20:18	US	192.168.16.99	1
16	mcouliba	2022-05-11	06:44:22	CAN	192.168.172.189	1
17	pwashing	2022-05-11	02:33:02	USA	192.168.81.89	1
18	pwashing	2022-05-11	19:28:50	US	192.168.66.142	0
19	jhill	2022-05-12	13:09:04	US	192.168.142.245	1
20	tshah	2022-05-12	18:56:36	MEXICO	192.168.109.50	0
21	iuduike	2022-05-11	17:50:00	US	192.168.131.147	1
22	rjensen	2022-05-11	00:59:26	MEX	192.168.213.128	0
23	yappiah	2022-05-10	18:11:53	MEXICO	192.168.200.48	1
27	aalonso	2022-05-10	01:55:35	MEX	192.168.103.210	0
29	bisles	2022-05-11	01:21:22	US	192.168.85.186	0
31	acook	2022-05-12	17:36:45	CANADA	192.168.58.232	0
33	zbernal	2022-05-11	02:52:10	US	192.168.72.59	1
34	drosas	2022-05-11	21:02:04	US	192.168.45.93	0
35	tshah	2022-05-10	15:26:08	MEX	192.168.92.147	0
37	eraab	2022-05-10	06:03:41	CANADA	192.168.152.148	0
40	aalonso	2022-05-12	15:15:46	MEX	192.168.174.186	0
41	apatel	2022-05-10	17:39:42	CANADA	192.168.46.207	0

Figure 1 — Login attempts strictly after 2022-05-09 (125 rows).

Logins On or After 2022-05-09 (>= operator)

```
SELECT *
FROM log_in_attempts
WHERE login_date >= '2022-05-09';
```

```
MariaDB [organization]> SELECT *
->
-> FROM log_in_attempts
->
-> WHERE login_date >= '2022-05-09';
```

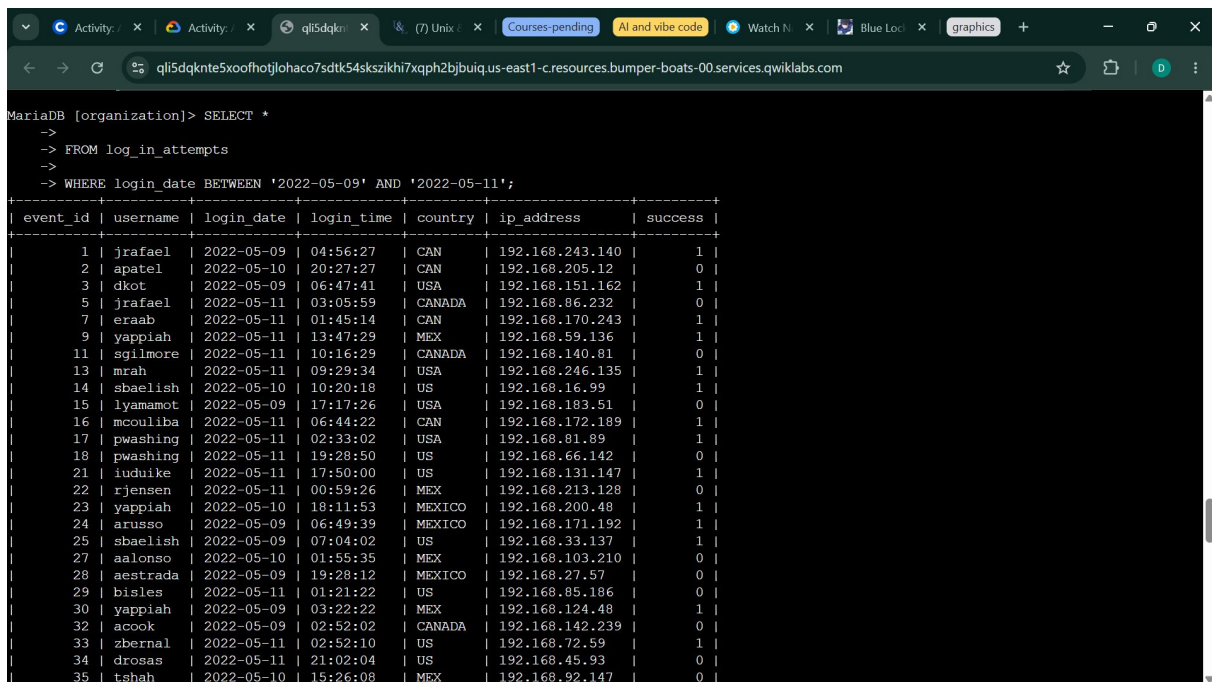
event_id	username	login_date	login_time	country	ip_address	success
1	jrafael	2022-05-09	04:56:27	CAN	192.168.243.140	1
2	apatel	2022-05-10	20:27:27	CAN	192.168.205.12	0
3	dkot	2022-05-09	06:47:41	USA	192.168.151.162	1
5	jrafael	2022-05-11	03:05:59	CANADA	192.168.86.232	0
6	arutley	2022-05-12	17:00:59	MEXICO	192.168.3.24	0
7	eraab	2022-05-11	01:45:14	CAN	192.168.170.243	1
9	yappiah	2022-05-11	13:47:29	MEX	192.168.59.136	1
10	jrafael	2022-05-12	09:33:19	CANADA	192.168.228.221	0
11	sgilmore	2022-05-11	10:16:29	CANADA	192.168.140.81	0
13	mrhah	2022-05-11	09:29:34	USA	192.168.246.135	1
14	sbaelish	2022-05-10	10:20:18	US	192.168.16.99	1
15	lyamamoto	2022-05-09	17:17:26	USA	192.168.183.51	0
16	mcouliba	2022-05-11	06:44:22	CAN	192.168.172.189	1
17	pwashing	2022-05-11	02:33:02	USA	192.168.81.89	1
18	pwashing	2022-05-11	19:28:50	US	192.168.66.142	0
19	jhill	2022-05-12	13:09:04	US	192.168.142.245	1
20	tshah	2022-05-12	18:56:36	MEXICO	192.168.109.50	0
21	iuduike	2022-05-11	17:50:00	US	192.168.131.147	1
22	rjensen	2022-05-11	00:59:26	MEX	192.168.213.128	0
23	yappiah	2022-05-10	18:11:53	MEXICO	192.168.200.48	1
24	arusso	2022-05-09	06:49:39	MEXICO	192.168.171.192	1
25	sbaelish	2022-05-09	07:04:02	US	192.168.33.137	1
27	aalonso	2022-05-10	01:55:35	MEX	192.168.103.210	0
28	aestrada	2022-05-09	19:28:12	MEXICO	192.168.27.57	0
29	bisles	2022-05-11	01:21:22	US	192.168.85.186	0
30	yappiah	2022-05-09	03:22:22	MEX	192.168.124.48	1

Figure 2 — Login attempts on or after 2022-05-09 (165 rows).

Task 2 — Retrieve Logins in a Date Range

To narrow the investigation window, the **BETWEEN ... AND** operator was used to retrieve login attempts that occurred between **2022-05-09** and **2022-05-11** inclusive. The query returned **123 rows**, excluding any attempts made after 2022-05-11.

```
SELECT *
FROM log_in_attempts
WHERE login_date BETWEEN '2022-05-09' AND '2022-05-11';
```



The screenshot shows a web browser window with a terminal interface for MariaDB. The terminal displays the following SQL query and its results:

```
MariaDB [organization]> SELECT *
->
-> FROM log_in_attempts
->
-> WHERE login_date BETWEEN '2022-05-09' AND '2022-05-11';
```

event_id	username	login_date	login_time	country	ip_address	success
1	jrafael	2022-05-09	04:56:27	CAN	192.168.243.140	1
2	apatel	2022-05-10	20:27:27	CAN	192.168.205.12	0
3	dkot	2022-05-09	06:47:41	USA	192.168.151.162	1
5	jrafael	2022-05-11	03:05:59	CANADA	192.168.86.232	0
7	eraab	2022-05-11	01:45:14	CAN	192.168.170.243	1
9	yappiah	2022-05-11	13:47:29	MEX	192.168.59.136	1
11	sgilmore	2022-05-11	10:16:29	CANADA	192.168.140.81	0
13	mrh	2022-05-11	09:29:34	USA	192.168.246.135	1
14	sbaelish	2022-05-10	10:20:18	US	192.168.16.99	1
15	lyamamot	2022-05-09	17:17:26	USA	192.168.183.51	0
16	mcouliba	2022-05-11	06:44:22	CAN	192.168.172.189	1
17	pwashing	2022-05-11	02:33:02	USA	192.168.81.89	1
18	pwashing	2022-05-11	19:28:50	US	192.168.66.142	0
21	iuduke	2022-05-11	17:50:00	US	192.168.131.147	1
22	rjensen	2022-05-11	00:59:26	MEX	192.168.213.128	0
23	yappiah	2022-05-10	18:11:53	MEXICO	192.168.200.48	1
24	arusso	2022-05-09	06:49:39	MEXICO	192.168.171.192	1
25	sbaelish	2022-05-09	07:04:02	US	192.168.33.137	1
27	aalonso	2022-05-10	01:55:35	MEX	192.168.103.210	0
28	aestrada	2022-05-09	19:28:12	MEXICO	192.168.27.57	0
29	bisles	2022-05-11	01:21:22	US	192.168.85.186	0
30	yappiah	2022-05-09	03:22:22	MEX	192.168.124.48	1
32	acook	2022-05-09	02:52:02	CANADA	192.168.142.239	0
33	zbernal	2022-05-11	02:52:10	US	192.168.72.59	1
34	drosas	2022-05-11	21:02:04	US	192.168.45.93	0
35	tshah	2022-05-10	15:26:08	MEX	192.168.92.147	0

Figure 3 — Login attempts between 2022-05-09 and 2022-05-11 (123 rows).

Task 3 — Investigate Logins at Certain Times

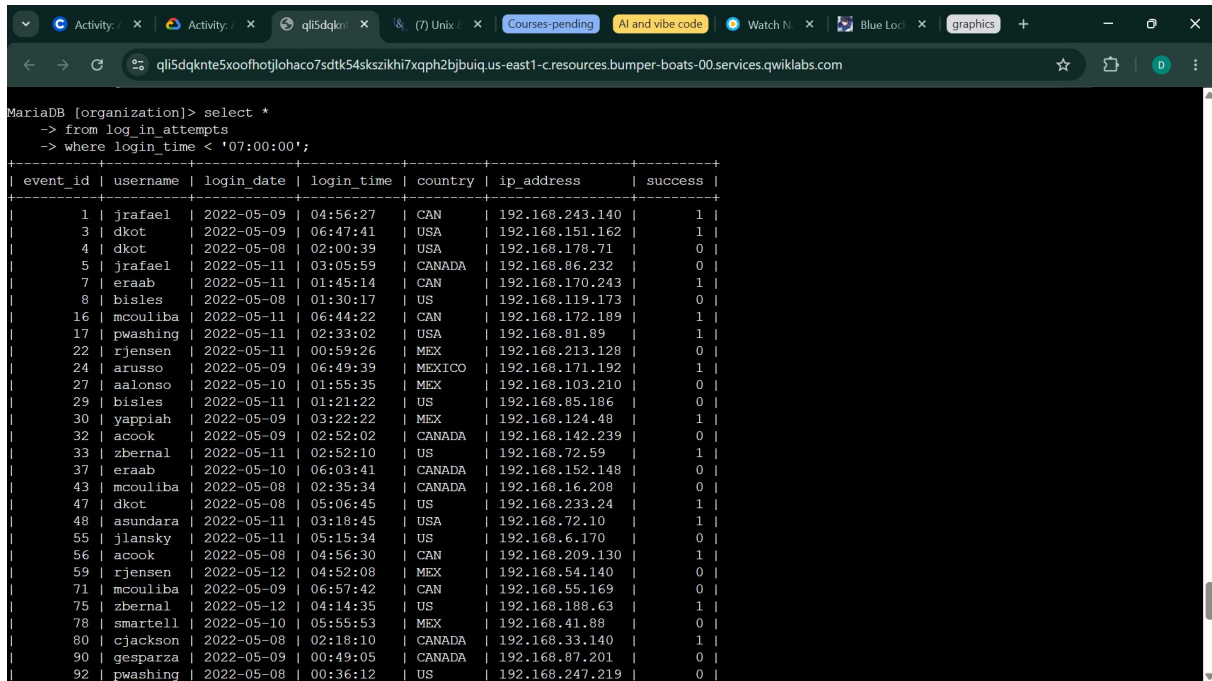
The organisation's standard working hours begin at 07:00:00. Login attempts made outside these hours were investigated to identify potentially suspicious activity.

Logins Before 07:00:00

A filter using the less-than operator on **login_time** returned all attempts made before 07:00:00. The fifth record in the result set belonged to user **eraab**. The query returned **67**

rows in total.

```
SELECT *
FROM log_in_attempts
WHERE login_time < '07:00:00';
```



The screenshot shows a web browser window with a terminal interface for MariaDB. The terminal displays a SQL query and its results. The query filters for login attempts before 07:00:00. The results are shown in a table with 7 columns: event_id, username, login_date, login_time, country, ip_address, and success. The 5th record is highlighted, showing a successful login for user 'eraab' at 06:01:31 from Canada.

event_id	username	login_date	login_time	country	ip_address	success
1	jrafael	2022-05-09	04:56:27	CAN	192.168.243.140	1
3	dkot	2022-05-09	06:47:41	USA	192.168.151.162	1
4	dkot	2022-05-08	02:00:39	USA	192.168.178.71	0
5	jrafael	2022-05-11	03:05:59	CANADA	192.168.86.232	0
7	eraab	2022-05-11	06:01:31	CAN	192.168.170.243	1
8	bisles	2022-05-08	01:30:17	US	192.168.119.173	0
16	mcouliba	2022-05-11	06:44:22	CAN	192.168.172.189	1
17	pwashing	2022-05-11	02:33:02	USA	192.168.81.89	1
22	rjensen	2022-05-11	00:59:26	MEX	192.168.213.128	0
24	arusso	2022-05-09	06:49:39	MEXICO	192.168.171.192	1
27	aalonso	2022-05-10	01:55:35	MEX	192.168.103.210	0
29	bisles	2022-05-11	01:21:22	US	192.168.85.186	0
30	yappiah	2022-05-09	03:22:22	MEX	192.168.124.48	1
32	acook	2022-05-09	02:52:02	CANADA	192.168.142.239	0
33	zbernal	2022-05-11	02:52:10	US	192.168.72.59	1
37	eraab	2022-05-10	06:03:41	CANADA	192.168.152.148	0
43	mcouliba	2022-05-08	02:35:34	CANADA	192.168.16.208	0
47	dkot	2022-05-08	05:06:45	US	192.168.233.24	1
48	asundara	2022-05-11	03:18:45	USA	192.168.72.10	1
55	jlansky	2022-05-11	05:15:34	US	192.168.6.170	0
56	acook	2022-05-08	04:56:30	CAN	192.168.209.130	1
59	rjensen	2022-05-12	04:52:08	MEX	192.168.54.140	0
71	mcouliba	2022-05-09	06:57:42	CAN	192.168.55.169	0
75	zbernal	2022-05-12	04:14:35	US	192.168.188.63	1
78	smartell	2022-05-10	05:55:53	MEX	192.168.41.88	0
80	cjackson	2022-05-08	02:18:10	CANADA	192.168.33.140	1
90	gesparza	2022-05-09	00:49:05	CANADA	192.168.87.201	0
92	pwashing	2022-05-08	00:36:12	US	192.168.247.219	0

Figure 4 — Login attempts before 07:00:00 (67 rows; 5th record: eraab).

Logins Between 06:00:00 and 07:00:00

The query was refined using **BETWEEN** to focus specifically on the 06:00–07:00 window. This returned **15 rows**. The earliest login attempt in this window was recorded at **06:01:31** by user lyamamot (event_id 182).

```
SELECT *
FROM log_in_attempts
WHERE login_time BETWEEN '06:00:00' AND '07:00:00';
```

```

187 | arusso | 2022-05-09 | 00:36:26 | MEX | 192.168.77.137 | 0 |
188 | jsoto | 2022-05-11 | 00:39:09 | USA | 192.168.21.88 | 0 |
189 | nmason | 2022-05-08 | 05:37:24 | CANADA | 192.168.168.117 | 1 |
190 | jsoto | 2022-05-09 | 05:09:21 | USA | 192.168.25.60 | 0 |
191 | cjackson | 2022-05-08 | 06:46:07 | CANADA | 192.168.7.187 | 0 |
195 | alevitsk | 2022-05-11 | 06:59:13 | CANADA | 192.168.236.78 | 1 |
200 | jclark | 2022-05-12 | 01:11:45 | CANADA | 192.168.91.103 | 1 |
67 rows in set (0.001 sec)

MariaDB [organization]> select *
-> from log_in_attempts
-> where login_time between '06:00:00' and '07:00:00';
+-----+-----+-----+-----+-----+-----+-----+
| event_id | username | login_date | login_time | country | ip_address | success |
+-----+-----+-----+-----+-----+-----+-----+
| 3 | dkot | 2022-05-09 | 06:47:41 | USA | 192.168.151.162 | 1 |
| 16 | mcouliba | 2022-05-11 | 06:44:22 | CAN | 192.168.172.189 | 1 |
| 24 | arusso | 2022-05-09 | 06:49:39 | MEXICO | 192.168.171.192 | 1 |
| 37 | eraab | 2022-05-10 | 06:03:41 | CANADA | 192.168.152.148 | 0 |
| 71 | mcouliba | 2022-05-09 | 06:57:42 | CAN | 192.168.55.169 | 0 |
| 98 | gesparza | 2022-05-11 | 06:30:14 | CANADA | 192.168.148.80 | 0 |
| 106 | tmitchel | 2022-05-12 | 06:15:41 | MEXICO | 192.168.3.252 | 1 |
| 134 | iuduke | 2022-05-09 | 06:46:40 | USA | 192.168.22.115 | 1 |
| 136 | mabadi | 2022-05-10 | 06:56:44 | US | 192.168.214.234 | 1 |
| 142 | gesparza | 2022-05-11 | 06:31:14 | CANADA | 192.168.117.56 | 1 |
| 147 | yappiah | 2022-05-08 | 06:04:34 | MEX | 192.168.65.245 | 0 |
| 148 | daquino | 2022-05-08 | 06:15:55 | CANADA | 192.168.135.6 | 1 |
| 182 | lyamamot | 2022-05-10 | 06:01:31 | USA | 192.168.106.52 | 0 |
| 191 | cjackson | 2022-05-08 | 06:46:07 | CANADA | 192.168.7.187 | 0 |
| 195 | alevitsk | 2022-05-11 | 06:59:13 | CANADA | 192.168.236.78 | 1 |
15 rows in set (0.001 sec)

MariaDB [organization]>

```

Figure 5 — Login attempts between 06:00:00 and 07:00:00 (15 rows; earliest: 06:01:31).

Task 4 — Investigate Logins by Event ID

The final task filtered login attempts by numeric **event_id**, selecting only the **event_id**, **username**, and **login_date** columns to reduce output noise. Numeric values were not enclosed in quotation marks.

Event IDs Greater Than or Equal to 100

Using the **>=** operator, all login events from event_id 100 onwards were retrieved. The third record returned showed a login date of **2022-05-09** (event_id 102, user jreckley).

```

SELECT event_id, username, login_date
FROM log_in_attempts
WHERE event_id >= 100;

```

MariaDB [organization]> select event_id, username, login_date
-> from log_in_attempts
-> where event_id >= 100;

event_id	username	login_date
100	tmitchel	2022-05-12
101	sbaelish	2022-05-08
102	jreckley	2022-05-09
103	jhill	2022-05-11
104	asundara	2022-05-11
105	cjackson	2022-05-12
106	tmitchel	2022-05-12
107	bisles	2022-05-12
108	daquino	2022-05-09
109	mcouliba	2022-05-10
110	mabadi	2022-05-09
111	astrada	2022-05-10
112	rjensen	2022-05-09
113	gesparza	2022-05-10
114	smartell	2022-05-10
115	ivelasco	2022-05-10
116	tmitchel	2022-05-10
117	bsand	2022-05-08
118	smartell	2022-05-12
119	tmitchel	2022-05-11
120	tmitchel	2022-05-09
121	btang	2022-05-10
122	yappiah	2022-05-11
123	bmoreno	2022-05-10
124	asundara	2022-05-12
125	bisles	2022-05-11
126	jrafael	2022-05-12
127	abellmas	2022-05-09

Figure 6 — Login events with event_id >= 100 (3rd record: 2022-05-09).

Event IDs Between 100 and 150

The range was then narrowed using **BETWEEN 100 AND 150**. Counting down the result set, the seventh record was user **tmitchel** (event_id 106, login date 2022-05-12).

```
SELECT event_id, username, login_date
FROM log_in_attempts
WHERE event_id BETWEEN 100 AND 150;
```

```

MariaDB [organization]> select event_id, username, login_date
-> from log_in_attempts
-> where event_id between 100 and 150;

```

event_id	username	login_date
100	tmitchel	2022-05-12
101	sbaelish	2022-05-08
102	jreckley	2022-05-09
103	jhill	2022-05-11
104	asundara	2022-05-11
105	cjackson	2022-05-12
106	tmitchel	2022-05-12
107	bisles	2022-05-12
108	daquino	2022-05-09
109	mcouliba	2022-05-10
110	mabadi	2022-05-09
111	astrada	2022-05-10
112	rjensen	2022-05-09
113	gesparza	2022-05-10
114	smartell	2022-05-10
115	ivelasco	2022-05-10
116	tmitchel	2022-05-10
117	bsand	2022-05-08
118	smartell	2022-05-12
119	tmitchel	2022-05-11
120	tmitchel	2022-05-09
121	btang	2022-05-10
122	yappiah	2022-05-11
123	bmoreno	2022-05-10
124	asundara	2022-05-12
125	bisles	2022-05-11
126	jrafael	2022-05-12
127	abellmas	2022-05-09

Figure 7 — Login events with event_id between 100 and 150 (7th record: tmitchel).

Summary

This lab demonstrated the use of numeric and date/time comparison operators to conduct targeted security investigations in SQL. Key results included: isolating 125 login attempts strictly after 2022-05-09 and expanding to 165 when the boundary date was included; narrowing the window to 123 attempts across a three-day range using BETWEEN; identifying 67 after-hours logins before 07:00:00, refined further to 15 within the 06:00–07:00 window with the earliest at 06:01:31; and pinpointing specific events by ID range, confirming user tmitchel as the seventh result in the 100–150 band. These skills are directly applicable to timeline reconstruction and anomaly detection during real-world security incident investigations.